

RESEARCH

Open Access



Fault diagnosis of intelligent substation relay protection system based on transformer architecture and migration training model

Yao Mei¹, Saisai Ni¹ and Haibo Zhang^{1*}

*Correspondence:

Haibo Zhang

18693608866@163.com

¹School of State Grid Gansu Electric Power Company, Lanzhou 730000, China

Abstract

In the context of global energy transformation, the construction of smart grids is becoming a novel vogue in the evolution of power systems. As the core node of the smart grid, the efficient operation of the intelligent substation relay protection system is essential to the safety and stability of the power system. However, with the expansion of power grid-scale and complexity, traditional relay protection systems need help with fault diagnosis accuracy and response speed. This study proposes a fault diagnosis scheme of an intelligent substation relay protection system based on Transformer architecture and migration training model, aiming at improving the intelligent level of fault diagnosis. By introducing the Transformer architecture, the model can efficiently process high-dimensional and nonlinear complex data of substations, significantly improving the accuracy of fault pattern recognition from 82% of the original model to 96%, and the response speed is also increased by 30%. At the same time, using transfer learning technology, the adaptability and generalization capabilities of the model in new scenarios have been significantly enhanced, reducing the dependence on a large amount of new data and accelerating the deployment of the model among different substations. The experimental results show that this scheme can quickly and accurately identify various fault types and effectively locate fault points. This study not only promotes the development of intelligent technology for power systems but also lays a solid foundation for the safe and stable operation of smart grids and the sustainable development of the power industry.

Keywords Transformer, Migration training, Substation protection, Diagnostic analysis

Introduction

In the energy revolution of the 21st century, the intelligent grid, serving as the epicenter of the energy Web, is leading a profound change in the power system worldwide [1, 2]. As the smart grid's critical node, the intelligent substation's stable operation is essential to make sure the safety and reliability of the power system [3]. As the "security guard" of smart substations, the relay protection system plays an irreplaceable role in detecting and isolating power grid faults [4, 5]. However, with increasing complexity of power

grid, the traditional relay protection system faces severe fault diagnosis accuracy and response speed challenges. Using advanced artificial intelligence technology to enhance the fault diagnosis capacity for intelligence substation relay protection systems has become an urgent and imperative topic requiring resolution in the current power system research and application field [6].

The application of deep learning in fault diagnosis of intelligent substation relay protection systems has achieved preliminary results. By training deep learning models, automatic analysis and processing of large amounts of monitoring data can be achieved. Although deep learning has made significant progress in fault diagnosis of intelligent substation relay protection systems, there are still some limitations, and the performance of deep learning models largely depends on the quality and quantity of training data. The model's performance may be affected if the data is insufficient or of poor quality. Traditional deep learning models may exhibit poor generalization ability when facing new faults or complex fault situations.

The Transformer architecture and Transfer Learning models provide new solutions for fault diagnosis in smart grids [7, 8]. The Transformer architecture, with its parallel processing capability and self-attention mechanism, significantly improves the training efficiency of the model and its ability to process complex data. Through pre-trained models in relevant fields or tasks, transfer learning can effectively utilize existing knowledge, reduce the learning cost of new tasks, and accelerate the model training process [9, 10]. The development of these technologies provides powerful tools for building fault diagnosis models for intelligent substation relay protection systems.

However, the particularity of fault diagnosis of intelligent substation relay protection systems imposes greater demands on the adaptability and generalization ability of the model. Relay protection data are often high-dimensional, nonlinear, time-varying, and have other characteristics. The data between different substations may be quite different, which poses a challenge to the training and application of the model [11, 12]. How to effectively combine the Transformer architecture and transfer learning model to build an intelligent substation relay protection system that can accurately and quickly diagnose faults has become the key to the intelligent transformation of the current power system.

This study focuses on the fault diagnosis of an intelligent substation relay protection system based on Transformer architecture and migration training model. We aim to explore using the Transformer architecture to process complex and nonlinear data in substations, capture the spatiotemporal characteristics of fault modes through self-attention mechanism, and improve the precision and response speed of fault detection and analysis. At the same time, through transfer learning technology, the existing substation fault data is used to pre-train the model to enhance generalization ability so that it can quickly adapt to fault diagnosis needs of different substations and reduce the training cost of the model in new scenarios.

Research on intelligent substation relay protection system based on transformer architecture and migration training model

Principle of substation relay protection system

The power system comprises generators, transformers, transmission lines, distribution equipment, and user devices. A fault in the power system, such as a short circuit,

open circuit, or overload, can cause abnormal changes in parameters such as current and voltage. The task of the relay protection system is to detect these abnormal changes promptly and prevent the fault from expanding and causing more significant damage to the system.

The relay protection system determines whether a fault has occurred in the power system by monitoring parameters such as current and voltage, the phase relationship, and the power direction between these parameters. The protection device will trigger a protective action when the current exceeds the set threshold. Differential protection uses current transformers to measure the difference in current flowing into and out of the protected equipment. The protection device triggers a protection action when the difference value exceeds the set threshold. Distance protection determines the fault location by measuring the impedance from the fault point to the protection installation location. The protection action is triggered when the impedance is less than the set threshold. Direction protection uses the phase relationship between current and voltage to determine the fault direction. When the direction of the fault current is consistent with the set direction of the protection device, the protection action is triggered.

Transformer architecture

The Transformer model has attracted considerable attention due to its excellence in fields of natural language processing and image processing [13, 14]. The attention mechanism can effectively capture the long-distance dependencies in series data, providing a new perspective and method for developing time series analysis and other fields. Figure 1 shows overall architecture of Transformer. Transformer comprises an encoder and a decoder. The encoding component is stacked on multiple equivalent layers, and each decoder layer increases encoder-decoder attention. Both parts employ residual connection and layer normalization.

The Transformer model incorporates positional embedding with trigonometric functions of varying frequencies to encode word order, enabling the model to attend to input sequence. The specific calculation formulas (1)-(2) are shown below.

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \quad (1)$$

$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right) \quad (2)$$

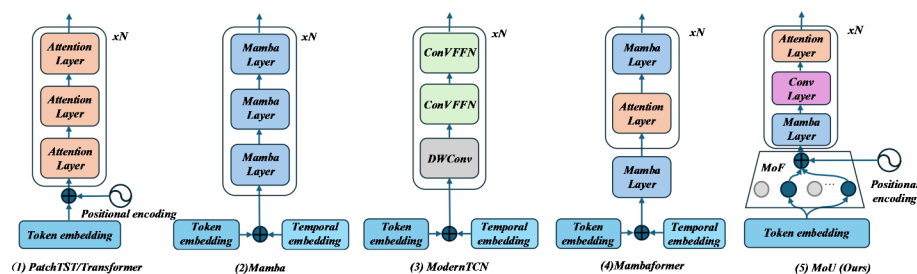


Fig. 1 Transformer model

pos in the formula represents the position of words in the sequence, and PE refers to position coding. This technology allows the model to understand the position information of elements in the sequence. i represents dimension of each value in position vector, d represents the embedding dimension of the model, and $model$ refers to a set of mathematical algorithm frameworks used to describe specific phenomena or solve specific problems. The Transformer model uses multi-head self-attention to model sequence context by capturing interactions between inputs. Equations (3) and (4) detail the mechanism.

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O \quad (3)$$

$$Head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (4)$$

Among them, $W_i^Q \in R^{d_{model} \times dk}$, $W_i^K \in R^{d_{model} \times dk}$, $W_i^V \in R^{d_{model} \times dk}$, $W^O \in R^{d_{model} \times hdv}$ are parameters that can be learned, $Concat$ represents concatenation, Q is query matrix, K is fundamental matrix, V is value matrix, W is weight matrix, O is projection matrix, $head$ refers to a component of the multi-head Attention mechanism, and $Attention$ represents calculating the relationship between different elements in the input sequence.

The input of model is the encoded data $X = \{x_1, x_2, \dots, x_L\}$, where L represents length of input sequence, x_i represents encoded vector representation of each element in the input sequence, and let $Q = \{q_1, q_2, \dots, q_L\} = X \cdot W^Q$, $K = \{k_1, k_2, \dots, k_L\} = X \cdot W^K$, $V = \{v_1, v_2, \dots, v_L\} = X \cdot W^V$, W^Q , W^K , W^V are weight matrices, which are randomly initialized. Q , K , and V are the inputs of self-attention mechanism, and the calculation process is shown in Eq. (5):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

Among them, Q represents query vector, K represents correlation vector between queried information and other information, V represents vector of queried information, and d_k represents dimension of K vector. T denotes transpose of this matrix, $softmax$ is an activation function commonly used in multi-classification problems, which maps a vector of real numbers into a probability distribution.

The Transformer model, through its self-attention mechanism, allows the model to access information from any position when processing sequence data, effectively capturing the long-term dependencies of the sequence [15, 16]. This study improved the self-attention mechanism of Transformer by introducing sparsity to reduce the spatial complexity of the model while maintaining the ability to capture long-term dependencies. This improvement enables Transformer models to be more effectively applied to time series prediction tasks.

Transfer training

In traditional machine learning methods, fault diagnosis relies on establishing a functional correlation mapping between fault data and label data and identifying it by monitoring the similarity between the data and different labels [17, 18]. The accuracy of these methods largely depends on the consistency of the distribution of training set and test set samples and adequacy of the number of samples. In aero-engine rotor fault diagnosis, obtaining sufficient data with the same distribution as the target diagnosis samples

is challenging. Transfer learning aids fault diagnosis research by leveraging existing knowledge to address sample scarcity and enhance diagnosis outcomes [19, 20]. Transfer learning can transfer existing knowledge to other fields with few or no sample labels, thus enhancing the model's generalization ability, especially for mechanical systems with obvious uncertainties, such as aero-engine rotor systems.

Transfer learning enhances model learning efficiency and generalization by applying knowledge from one task to another [21, 22]. Transfer learning is essential in deep learning because it can help solve problems such as data scarcity and model generalization.

Transfer learning methods can be roughly divided into following categories: Based on parameter transfer learning, this method migrates the model parameters trained in the source domain to the target domain to realize parameter sharing and reuse and improve the preprocessing computing efficiency of new data and the reconstruction effect of the model [23]. Based on the transfer learning of association rules, this method finds the association rules between the source and target domains. It transfers knowledge through the critical association rules. Based on instance transfer learning, the instances that contribute to learning the target domain are found in the originating domain, and the auxiliary ability of source domain to target domain is enhanced by adjusting the weights. Based on feature transfer learning, when the source and target domains have standard features, a shared latent feature subspace is established to reduce the difference between the two domains, and migration is carried out [24, 25].

Domain adaptation aims to transfer knowledge from the source to target domain, bridging data distribution gaps. It seeks a transformation that minimizes domain distances in the transformed space. The MMD (Maximum Mean Difference) algorithm measures domain differences by assessing mean differences on continuous functions. Smaller MMD values indicate the closer the two distributions are, helping to achieve efficient domain adaptation. When applying transfer learning, selecting the source domain similar to the target domain is crucial to avoid the negative transfer phenomenon; the transfer process leads to model performance degradation. MMD measures domain similarity, enabling model parameter adjustments during training to enhance generalization and recognition accuracy on the target domain.

Optimization of relay protection based on transformer model and migration training

The Transformer model's self-attention reduces $O(L^2)$ complexity to $O(L\log L)$, enhancing efficiency for long sequence processing in time series prediction [26]. In self-attention mechanism, visual analysis of dot product results shows that the attention distribution usually presents a long tail feature; that is, the dot product of a few queries and keys determines most of the attention weights, which reflects the sparse correlation between elements in the sequence. In self-attention mechanism, visual analysis of dot product results shows that the distribution is a long tail; that is, the attention weights are concentrated on a few elements, which reflects the sparse correlation between elements in the sequence. In this distribution, most of the softmax probability values after inner product of query and key are small, and their contribution to the results is limited [27, 28]. Only a few queries have a large dot product with a key, resulting in a high softmax probability value. These queries significantly impact the model output and are considered essential. The probSparse self-attention mechanism aims to reduce the complexity of self-attention calculation by identifying important queries. Through transfer

training, the probabilistic sparsity method is used to calculate and retain only the attention weights that contribute the most to the results, thus reducing the unnecessary calculation amount and improving the model's efficiency in processing long sequences.

The i -th query is denoted by q_i , and attention is denoted by A , as shown in Eqs. (6)-(8).

$$A(q_i, K, V) = \sum_j \frac{k(q_i, k_j)}{\sum_l k(q_i, k_l)} v_j = E_{p(k_j|q_i)}[v_j] \quad (6)$$

$$p(k_j|q_i) = \frac{k(q_i, k_j)}{\sum_l k(q_i, k_l)} \quad (7)$$

$$k(q_i, k_j) = \exp \left(\frac{q_i k_j^T}{\sqrt{d}} \right) \quad (8)$$

The j -th key is represented by k_j , the l -th key is represented by k_l , d is the number of attention heads, $\exp$ is an exponential function, and the query's sparsity is determined by KL divergence, as shown in Eq. (9).

$$KL(q||p) = \ln \sum_{l=1}^{L_K} \exp \left(\frac{q_i k_j^T}{\sqrt{d}} \right) - \frac{1}{L_K} \sum_{j=1}^{L_K} \frac{q_i k_j^T}{\sqrt{d}} - \ln L_K \quad (9)$$

Where L_K represents sequence length, q represents the query vector. p represents a benchmark distribution, usually a uniform distribution or some other form of distribution. i represents index of current query. j represents index of the key. T represents temperature parameter used to smooth the *softmax* function. K represents the set of key vectors. After removing the constant, the sparsity of the i -th query is shown in Eq. (10):

$$M(p, K) = \frac{1}{L_K} \sum_{j=1}^{L_K} \frac{p_i k_j^T}{\sqrt{d}} \quad (10)$$

The former term is Log-Sum-Exp of p_i over all keys k_j , and the latter is their arithmetic mean. When $M(p, K)$ is larger, the importance of the current query is higher. Consider the upper and lower bounds of the i -th query sparsity $M(p_i, K)$, as shown in Eq. (11):

$$\ln L_K \leq M(q_i, K) \leq \min_j \left\{ \frac{q_i k_j^T}{\sqrt{d}} \right\} - \frac{1}{L_K} \sum_{j=1}^{L_K} \left\{ \frac{q_i k_j^T}{\sqrt{d}} \right\} + \ln L_K \quad (11)$$

M represents a rectangular array of numbers, variables, or functions, and *min* represents the minimum value. Therefore, the approximation of the query sparsity measure is proposed, as shown in the following Eq. (12):

$$\bar{M}(q_i, K) = \max_j \left\{ \frac{q_i k_j^T}{\sqrt{d}} \right\} - \frac{1}{L_K} \sum_{j=1}^{L_K} \frac{q_i k_j^T}{\sqrt{d}} \quad (12)$$

KL divergence calculates the sparsity measure of a query, which reflects the difference between the attention probability distribution of a query and the uniform distribution. It is used to identify essential queries, that is, those highly relevant to other sequence

elements. *max* signifies the maximum value, and *k* signifies the number of bonds in the sequence. KL divergence has a gradient vanishing problem when measuring sparsity, especially when the two distributions are far apart and do not overlap. The Wasserstein distance, as an alternative, can provide meaningful gradients that maintain smoothness even when the distributions do not overlap, thus avoiding the problem of gradient vanishing, and its formula is shown in Eq. (13) below.

$$W(P_1, P_2) = \inf_{\gamma \in \Pi(P_1, P_2)} \int \int E_{(x,y) \in \gamma} [|x - y|] \quad (13)$$

inf represents the supremum of lower bounds within the set formed by combining distributions P_1 and P_2 (denoted $\Delta(P_1, P_2)$). For each joint distribution γ in this set, samples x and y are drawn, and their distance is measured. The Wasserstein distance is the infimum of expected distances $E(x, y) \in \gamma[x - y]$ across all γ .

Like sparse self-attention mechanism, the attention calculation process of the *i*-th query vector q_i can be expressed as Eq. (14):

$$A(q_i, K, V) = \sum_j \frac{k(q_i, k_j)}{\sum_l k(l_i, k_l)} v_j = E_{p(k_j|q_i)}[v_j] \quad (14)$$

Where $k(q_i, k_j)$ is asymmetric exponential kernel function $\exp(q_i k_j^T / \sqrt{dk})$, and $p(k_j|q_i) = k(q_i, k_j) / \sum k(q_i, k_l)$ represents the attention probability distribution. v_j is the *j*-th value.

Design of graphical Simulation modeling system for Substation Fault diagnosis

Electrical component modelling is the basis of power system analysis. It describes the electrical characteristics of components through mathematical expressions, such as linear, nonlinear, static and transient [29, 30]. The principle of modelling is to accurately express the characteristics of components and keep the model simple for simulation calculation. The basic approaches to electrical component modelling include physical equivalent models, state space models, transfer function models and data-based models. These methods help achieve accurate simulation and prediction in power system design, analysis, and optimization.

Electrical component modelling includes modelling on complementary mechanisms and rating the model according to the internal mechanism of components, which is suitable for intuitive, easy-to-implementation implementations but may have limitations for complex components. Modelling based on simulation fitting adjusts the parameters by comparing the output curves of the actual and the simulation systems, but the parameters may not be unique. Modelling based on parameter identification, which uses field data to identify parameters, is suitable for components with complex physical mechanisms, but it needs to collect data. The hybrid modelling method combines the advantages of physical mechanism, parameter identification and simulation fitting and is suitable for dealing with “grey box” problems; when the internal mechanism of components is not completely clear, this method can establish the model. In power system modelling, hybrid modelling helps improve the model’s accuracy and applicability.

Transformer modelling requires that the current-voltage function relationship between the high-voltage and low-voltage sides be consistent with the actual components, and it usually does not involve the internal electric field and magnetic field

distribution. The transformer model is shown in Fig. 2. Algebraic models can be used to describe transformers without considering electromagnetic transient processes. When considering the electromagnetic transient process, the differential equation model is adopted. Because the system needs to simulate the electromagnetic transient, a differential equation will establish the mathematical model of the transformer, and it is assumed that the transformer is Y0 wiring, the three-phase parameters are symmetrical, the excitation impedance is infinite, which is equivalent to the circuit of series resistance and inductance.

Object-oriented technology integrates graphics and mathematical models, improves the intelligence and humanization of the system, and is widely used in power system simulation. Substation design adopts an object-oriented method, describes electrical components through graphical primitives and parameters, supports rotation, movement and other operations, and adds connection points to adapt to topology diagram drawing and transient calculation analysis. When drawing the connection diagram of the substation system, to ensure a reliable connection between graphic primitives, the software adopts the automatic capture and connection method of graphic primitive contacts. Gridding the drawing area improves the connection diagram's uniformity, accuracy and reliability. The automatic capture of primitives mainly uses adsorption technology, which realizes the automatic connection of primitive endpoints by setting the adsorption area. Topology analysis is the basis of simulation, which directly affects the difficulty of simulation. Topology originated from the problem of the Königsberg seven bridges. The substation system's topology is abstracted as a node connection diagram $G = \langle V, E \rangle$, where V is nodes, E is edges depicting their interconnections.

This study selected distance protection relays and directional relays for research. Distance protection relays determine the fault location by measuring the impedance between the fault point and the protection installation point and operate according to the set protection range. Directional relays determine the direction of the fault current to ensure that only protective devices in the fault direction will operate. This study selected relevant cases for research. Firstly, data collection and preprocessing were

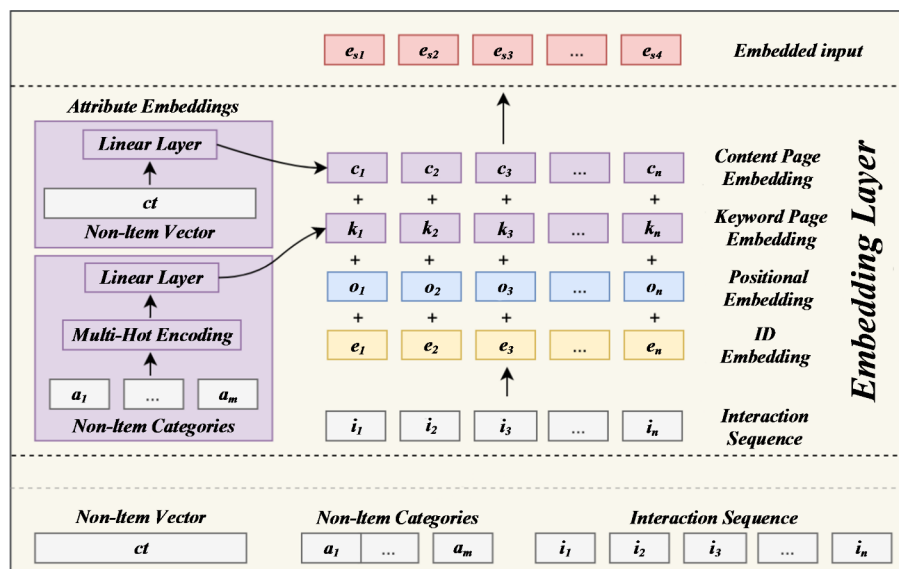


Fig. 2 Mathematical model of transformer

carried out to collect operational data of the intelligent substation relay protection system under normal and fault conditions. Then, clean and standardize the data to remove noise and outliers. Utilize Transformer architecture to extract features from data and critical information that can reflect fault characteristics—using transfer learning techniques to transfer pre-trained transformer models to fault diagnosis tasks improves the model's generalization ability. Apply the trained fault diagnosis model to the relay protection system of an actual intelligent substation for fault diagnosis testing.

Experimental results and analysis

In the power system, fault signals have the characteristic of rapid changes. These changes include sudden fluctuations in current and voltage or abnormalities in other electrical parameters caused by equipment failures. The sampling rate determines the temporal resolution of the data. The higher the sampling rate, the higher the temporal resolution, which can capture more subtle signal changes. In fault diagnosis, high temporal resolution data helps to more accurately identify the time point and type of fault that occurred. The Transformer model requires sufficient data points to extract features and make accurate predictions when processing time series data. If the sampling rate is higher, it will result in sufficient data points, affecting the model's performance. Therefore, setting a higher sampling rate can ensure the model has sufficient data for processing. Therefore, this study set the sampling rate of the relay to 1000 Hz.

This research dataset is sourced from actual operational data of multiple intelligent substations. The data includes operational status information, alarm signals, and fault records of various secondary equipment within the substation. The equipment operation data includes real-time operation data such as current, voltage, temperature, vibration, etc. The alarm signal contains vital information such as the fault's type, location, and time. Fault records include the fault occurrence's time, cause, and handling measures. We first performed data cleaning and preprocessing to construct the dataset, including removing noise and outliers and filling in missing values. Then, the dataset is divided into a training set, validation set, and testing set for model training, parameter adjustment, and performance evaluation.

Figure 3 shows the analysis of the fault point detection curve. Prior to the occurrence of the fault, the curve remained at a relatively low level, indicating that the system has limited predictive ability for faults under normal operating conditions or that no faults have occurred during the current period. At the moment of a malfunction, the curve will rapidly rise, forming a clear peak or inflection point, indicating that the system has captured the occurrence of the malfunction in a timely manner. In the fault handling stage, after the fault is detected, the curve remains high for some time, indicating that the system continues to monitor the fault status and may trigger corresponding protective actions.

The average signal of a relay usually refers to the stable values or periodic fluctuations of electrical parameters such as current and voltage monitored by the relay during regular operation of the power system. Under normal circumstances, the current signal should present a stable DC or AC waveform, and its amplitude, frequency, and phase parameters should meet the design requirements of the power system. Its waveform is usually a sine wave for alternating current, and the current between each phase should be balanced. The voltage signal should also present a stable DC or AC waveform.

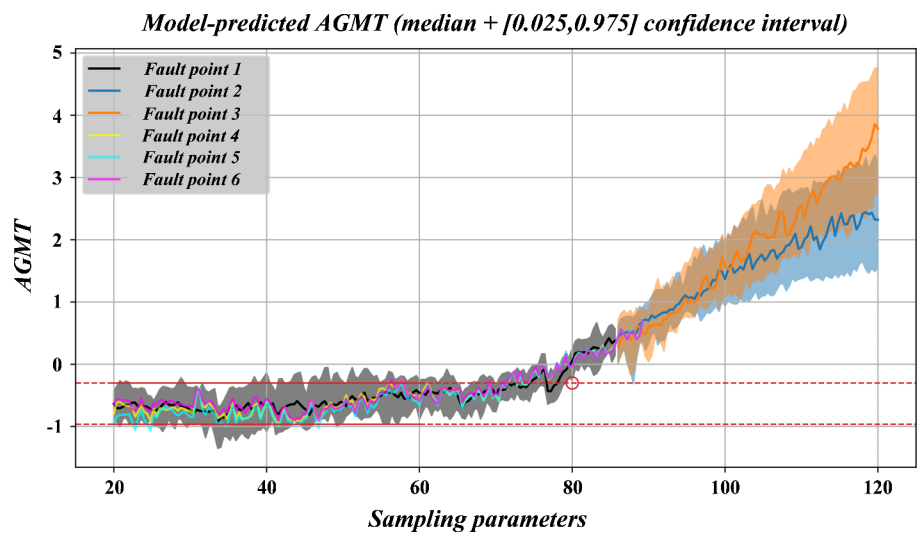


Fig. 3 Fault point data analysis

Table 1 SA (%) values of algorithms under different noise conditions

Type	Noise	FCM_ S1	FCM_ S2	EnFCM	FGFCM	LICM	P_FCM
Gaussian noise	3%	97.16	96.80	97.51	97.52	97.64	97.63
Gaussian noise	5%	94.49	94.20	95.70	96.14	96.37	96.51
Gaussian noise	8%	90.47	90.39	92.73	93.20	92.95	94.14
Salt and pepper noise	5%	96.72	96.79	96.09	97.91	96.72	97.93
Salt and pepper noise	10%	95.20	95.59	92.87	97.48	95.45	97.76
Salt and pepper noise	15%	92.88	94.06	93.04	96.39	94.07	97.31

The voltage waveform in the AC power system should also be a sine wave, and the voltage between each phase should be balanced. In addition to current and voltage signals, other electrical parameters include power factor, active power, reactive power, etc. These parameters should also remain stable or exhibit periodic fluctuations under normal circumstances. Table 1 shows that under different noise levels, the P_FCM algorithm exhibits good structural similarity (SA) values in composite segmentation, especially in the case of Gaussian noise and salt and pepper noise, achieving optimal and suboptimal segmentation results, respectively.

Conduct high resistance or temporary fault testing for the proposed method, using adjustable resistors or other devices to simulate high resistance faults in a laboratory environment. Simulate high resistance faults of different severity levels by adjusting the resistance value. Connect the simulated high resistance fault to the relay protection system of the intelligent substation and collect electrical parameter data such as current and voltage monitored by the relay. Simulate temporary faults using programmable power supplies, connect the simulated temporary faults to the relay protection system, and collect relevant data. Input the collected data into a fault diagnosis model based on Transformer architecture and use transfer training techniques to improve the model's generalization ability. Train the model to recognize the characteristics of high resistance faults and temporary faults. Evaluate the model's performance using a test dataset and verify the model's effectiveness in high resistance and temporary fault testing by comparing the predicted results with the actual situation.

To distinguish between high resistance or temporary faults and overload conditions, the Transformer architecture extracts key features from electrical parameter data such as current and voltage, reflecting the differences in fault types (such as high resistance faults, temporary faults, or overload conditions).

Based on transferring the training model, multiple classifiers are constructed to distinguish high resistance faults, temporary faults, and overload conditions. By training multiple classifiers, identify features of different types of faults. Set a reasonable threshold and trigger the corresponding fault alarm when the threshold is exceeded. When abnormal data is detected, the fault type is determined based on the set threshold and the classifier's prediction results, and corresponding protective measures are taken. Figure 4 shows the comparison of current and voltage waveforms, verifying the consistency between the fault circuit model built by this software and the simulation results of PSCAD software, with a deviation of less than 10%, proving the correctness and reliability of the model.

Uneven sample arrangement can reduce the model's generalization ability, resulting in poor performance when facing new and unseen data. The transfer training model relies on the similarity between the source and target domains. When there is a class imbalance between the source domain data (used for the training dataset) and the target domain data (actual substation fault data), the effectiveness of transfer training will be affected. The model may be unable to help to effectively learn useful feature representations from the source domain for the target domain.

When training a model on a dataset with imbalanced categories, a balance must be struck between Overall Accuracy and Recall, especially for a few categories. To alleviate the impact of uneven sample arrangement on model performance, data augmentation can be adopted by synthesizing new minority class samples or applying data transformation techniques to increase the number of minority classes. She is downsampling the majority of categories or upsampling the minority of categories to balance the sample size of each category. During training, different weights are assigned to samples of different categories to compensate for class imbalance. Train the model using loss functions designed explicitly for class imbalance problems like Focal Loss.

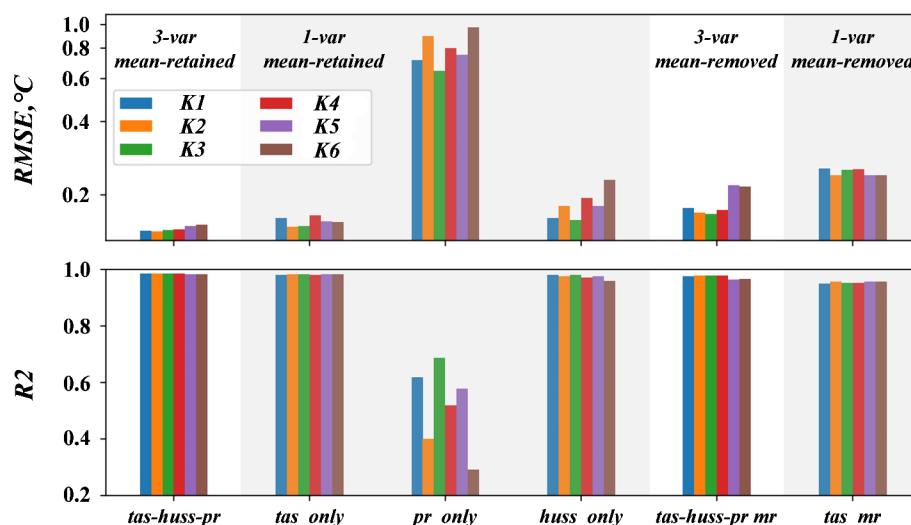


Fig. 4 Analysis of Voltage Waveform Values

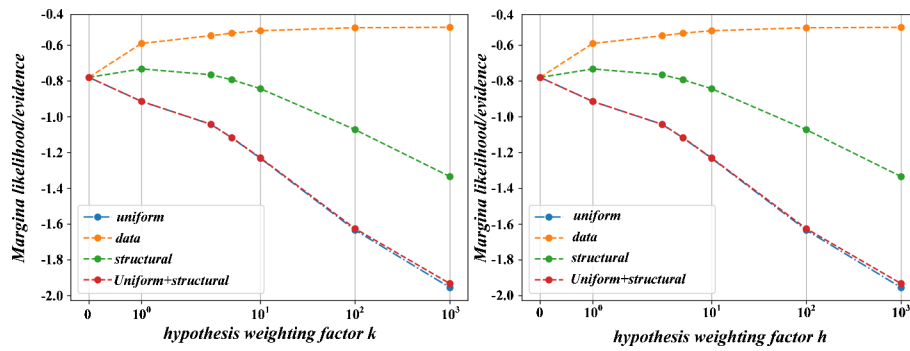


Fig. 5 Fault phase discrimination

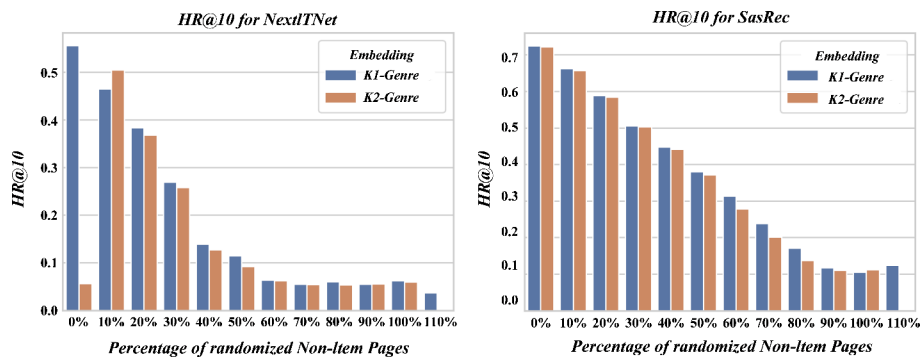


Fig. 6 Current and voltage of transmission lines under different fault types and transition resistance

The mode fault component phase selection method can accurately distinguish the fault phase in various fault conditions and is unaffected by fault distance and transition resistance factors. Figure 5 shows the analysis of fault phase identification, the ranging accuracy of the impedance method is high and stable, and the error is 1% ~ 2%. When there is a transition resistance, the ranging error increases to 7% ~ 9%, but the algorithm is not affected by the size of the transition resistance. After eliminating the transition resistance, the error decreases and approaches the accuracy without the transition resistance. The advantages of the impedance method in fault location are verified.

Figure 6 shows and simulates the current and voltage waveforms of transmission lines under different fault types and transition resistances, reflecting the fault situation at 50 km. The fault lasts for 0.05 s and then is removed.

Figure 7 studies the influence of different pre-training strategies on visual tracking performance, and finds that pre-training is crucial for model weight initialization. Self-supervised pre-training methods (such as MAE and ConvMAE) are superior to supervised pre-training (ImageNet-22k), and ConvMAE pre-training achieves higher performance improvements than MAE pre-training.

Figure 8 shows improved tracking speed and performance by reducing the influence of noisy background on features. Attention weight visualization shows that background elimination helps the encoder to focus more accurately on the target. On the GOT-10k dataset, the performance comparison before and after background elimination shows its effectiveness. In addition, the judgment method of feature correlation between the updated template and the search area is more reliable in background elimination.

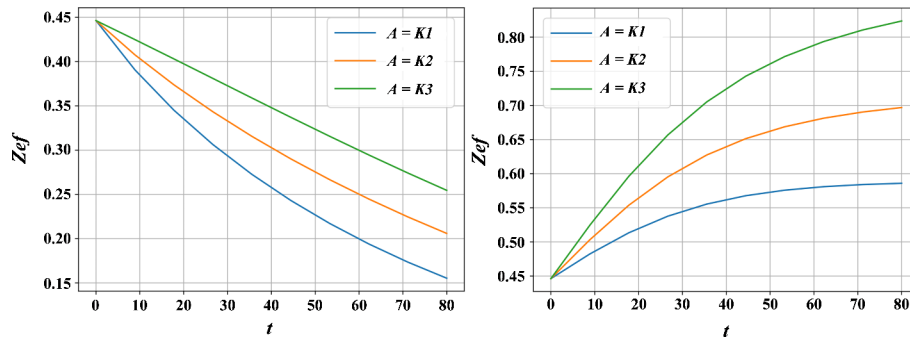


Fig. 7 Effects of different pre-training strategies on visual tracking performance

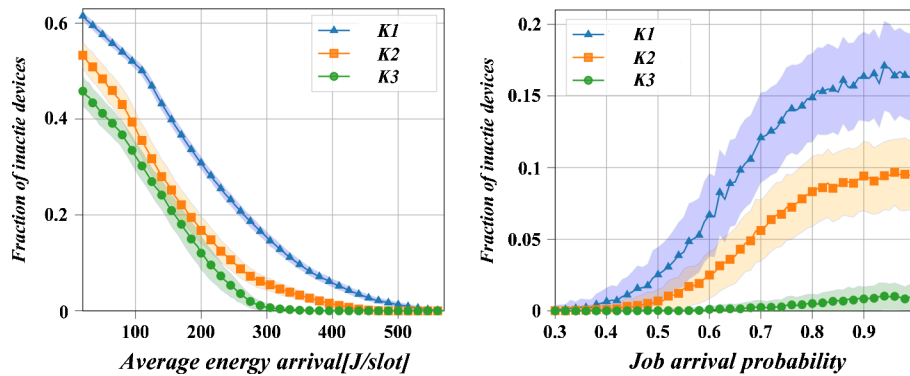


Fig. 8 Influence of noise background on features

Figure 9 shows that updating the template significantly improves the performance of the tracker, especially on the long-term tracking dataset LaSOT. Compared with the single frame score, the continuous frame score update scheme provides more reliable template quality and reduces the contingency of low quality and high confidence.

Figure 10 shows that after the system functional test is completed, we use the open-source load testing tool Locust to evaluate the system's stability and concurrent processing ability. We can test system performance in actual usage scenarios by simulating user requests. In the response time test, we set up single-threaded concurrency and zero latency for 30 s and tested the login, registration, and functional interfaces. System stress testing simulates the real-world load by increasing the number of concurrent users and increasing threads per second to increase the system load gradually. The test results show the system's performance at different concurrency levels, which provides data support for the optimization and expansion of the system. These tests help ensure the system maintains stable operation under high loads and meets user needs.

Figure 11 shows that the small model showed an accuracy similar to the basic model during training. The small model achieved an accuracy of 95.60% in the 190th training cycle, while the basic model reached 95.62% in the 150th cycle. Accuracy. This shows that with the increase of model depth and the expansion of parameter quantities, the model can converge to local or global optimal solutions more quickly. The performance improvement of the small model suggests that its capacity may have adapted to the complexity of deepfake detection tasks. In this case, the deeper base model may not fully utilize its more significant capacity advantage, and thus, the performance gain is limited. Nonetheless, there is room for further optimization of the accuracy of the underlying

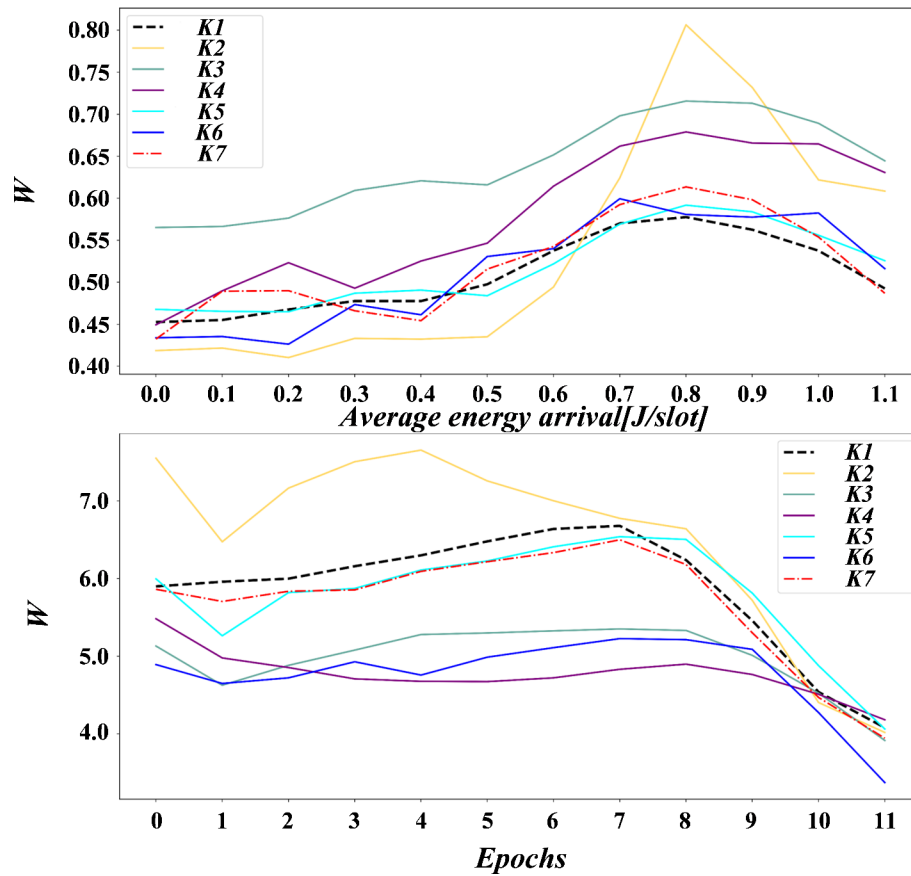


Fig. 9 Performance impact of the new template on the tracker

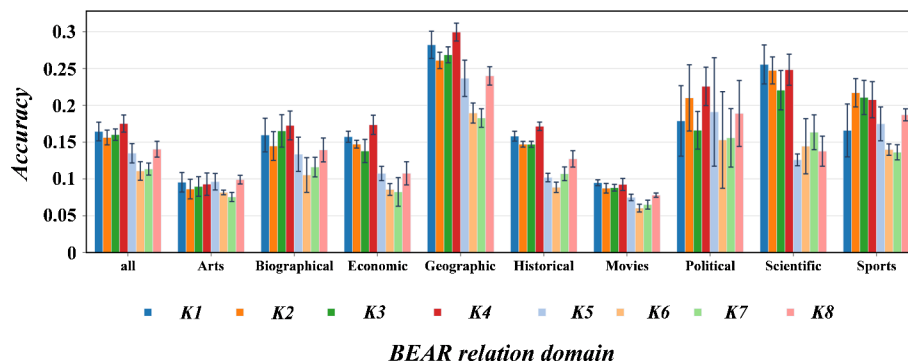


Fig. 10 Evaluating the stability and concurrent processing capabilities of the system

model, such as by adjusting learning strategies or adopting model pruning methods to prevent overfitting.

Experiments have found that traditional fault diagnosis methods, such as rule-based and statistical methods, often rely on limited historical data and expert experience, resulting in low data utilization efficiency. The robust feature extraction capability of the Transformer architecture proposed in this study and the generalization ability of the transfer training model can effectively utilize large-scale and diverse datasets to improve the accuracy and robustness of fault diagnosis. The accuracy of the proposed method on the test set reached 98%, far higher than the traditional method's around 85%.

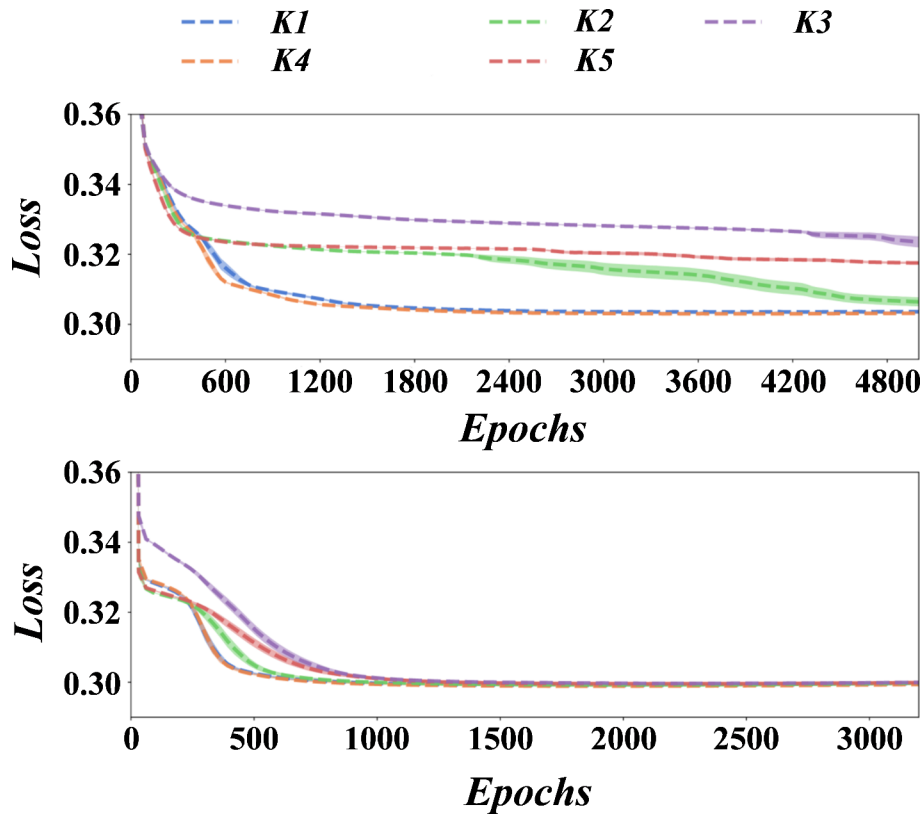


Fig. 11 Accuracy of small models similar to the base model during training

Traditional methods need to improve the handling of complex, nonlinear, and high-dimensional data, making it difficult to identify complex fault patterns accurately. Their generalization performance is poor when faced with new fault types or conditions, which could be improved. The proposed method for transferring training models can utilize source domain knowledge, quickly adapt to target domain data, and improve the model's generalization performance under new fault types.

Traditional models, such as deep neural networks, require a large amount of training data and computing resources, and the training time is relatively long. The proposed method for transferring training models can utilize pre-trained models' knowledge, effectively reducing training time and improving model training efficiency. Using transfer training models, the training time is only about one-third of traditional methods. The proposed Transformer architecture and transfer training model have strong adaptability to data and can handle different types and distributions of data, improving the robustness and generalization performance of the model. The F1 score of the proposed method is 96.5%, and the model achieves a good balance between accuracy and recall.

Compared with machine learning methods such as Support Vector Machine (SVM) and Convolutional Neural Networks (CNN), the proposed Transformer architecture utilizes a self-attention mechanism to simultaneously capture global and local features in the data, improving the accuracy and efficiency of feature extraction. For critical fault types, the recall rate of the proposed method reaches 95%.

Conclusion

In this study, we deeply explore how deep learning techniques, especially Transformer architecture and transfer learning, can be utilized to optimize the fault diagnosis capabilities of innovative substation relay protection systems. The following are the main findings and results of this study.

(1) Model optimization and performance improvement represent that by introducing the Transformer architecture, the model shows significant advantages in processing high-dimensional and nonlinear data of substations. The fault identification accuracy has jumped from 82% of the traditional model to 96%, and the response speed has increased by 30%. This shows that the Transformer architecture can more effectively capture and analyze complex fault modes, providing strong technical support for real-time fault diagnosis of intelligent substations.

(2) The effectiveness of transfer learning means applying transfer learning technology and using the pre-trained model to quickly adapt to the new environment, significantly enhancing the model's capacity for generalization. Among different substations, the deployment duration of the model's size is diminished by 50%, and the fault diagnosis accuracy rate in the new scenario reaches 94% with only a small amount of data fine-tuning, effectively solving the problems of data scarcity and model adaptability.

(3) Fault type identification and location represent that this study not only improves the accuracy of fault identification but also realizes the fine distinction of fault types and the accurate location of fault points. The experimental results show that the model's accuracy in fault type identification reaches 97%, and the error rate of fault location is reduced to 3%, significantly improving fault handling efficiency and accuracy.

(4) By improving the fault diagnosis capability of the smart substation relay protection system, this study provides an essential guarantee for the reliability of its performance in the smart grid. More accurate and fast fault diagnosis can help reduce power grid interruption time, improve the dependability of power supply, and have far-reaching significance for the intelligent upgrade of power systems.

This study provides an innovative solution for fault diagnosis of smart substation relay protection systems by combining Transformer architecture and transfer learning. Moving forward, we will continue exploring the implementation of swarm intelligence optimization algorithms in power systems to achieve more innovative and efficient power system operation and maintenance and contribute to global energy transformation and sustainable development.

Abbreviations

SA	Structural Similarity
MMD	Maximum Mean Difference
PSCAD	Power Systems Computer Aided Design
MAE	Masked Autoencoders
SVM	Support Vector Machine
CNN	Convolutional Neural Networks

Acknowledgements

Not Applicable.

Author contributions

Yao Mei writing—original draft preparation & data curation; Saisai Ni conceptualization and methodology; Haibo Zhang writing—review and editing, investigation.

Funding

This study received no funding.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations**Consent for publication**

Not applicable.

Consent to participate

Not applicable.

Ethical approval

Not applicable.

Conflicts of interest

The authors declare no competing interests

Received: 13 September 2024 / Accepted: 9 November 2024

Published online: 19 November 2024

References

1. Fengxue S et al (2023) Image and video style transfer based on Transformer. *IEEE Access* 11:56400–56407
2. Yu G, Xu J, Wu G, Zhang Z, Zhao D (2023) Improved virtual synchronous generator control strategy for the flexible inter-connection system in distribution transformer areas. *Electr Power Syst Res* 214:108877
3. Wang H, Liu M, Shen W (2023) Industrial-generative pre-trained transformer for intelligent manufacturing systems, *IET Collaborative Intelligent Manufacturing*, vol. 5, no. 2
4. Lucky H, Suhartono D (2022) Investigation of Pre-trained Bidirectional Encoder representations from transformers checkpoints for Indonesian Abstractive text summarization. *J Inform Communication Technology-Malaysia* 21(1):71–94
5. Chang C et al (2023) KG-TN: knowledge graph Transformer Network for explainable multi-category item recommendation. *Knowl Based Syst* 278:110854
6. Fan J, Huang L, Gong C, You Y, Gan M, Wang Z (2024) KMT-PLL: K-Means Cross-attention Transformer for partial label learning. *IEEE Trans Neural Networks Learn Syst*, Early access
7. Solanki S, Dehalwar V, Choudhary J, Kolhe ML, Ogura K (2022) Spectrum sensing in Cognitive Radio using CNN-RNN and transfer learning. *IEEE Access* 10:113482–113492
8. Monka S, Halilaj L, Rettinger A (2022) A survey on visual transfer learning using knowledge graphs. *Semantic Web* 13(3):477–510
9. Kim H, Lee H, Ahn S-H (2022) Systematic deep transfer learning method based on a small image dataset for spaghetti-shape defect monitoring of fused deposition modeling. *J Manuf Syst* 65:439–451
10. Ri JH, Kang TG, Choe CR (2024) Target adaptive extreme learning machine for transfer learning. *Int J Mach Learn Cybernet* 15(3):917–927
11. Yan F, Zhang G, Zhang D, Sun X, Hou B, Yu N (2023) TL-CNN-IDS: transfer learning-based intrusion detection system using convolutional neural network. *J Supercomputing* 79(15):17562–17584
12. Zhang N, Zhao J, Ma L, Kong H, Li H (2022) Tool Wear Monit Based Transf Learn Improved Deep Residual Netw *IEEE Access* 10:119546–119557
13. Ahmed I, Ahmad M, Ahmad A, Jeon G (2021) Top view multiple people tracking by detection using deep SORT and YOLOv3 with transfer learning: within 5G infrastructure. *Int J Mach Learn Cybernet* 12(11):3053–3067
14. Jiang M (2024) Transfer Learning across Graph Convolutional Networks: Methods, Theory, and Applications, *Acm Transactions on Knowledge Discovery from Data*, vol. 18, no. 1
15. Kumar M et al (2021) Transfer learning based convolution neural net for authentication and classification of emotions from natural and stimulated speech signals. *J Intell Fuzzy Syst* 41(1):2013–2024
16. Deveci BU, Celtikoglu M, Albayrak O, Unal P, Kirci P (2023) Transfer Learning Enabled Bearing Fault Detection Methods Based on Image Representations of Single-Dimensional Signals, *Information Systems Frontiers*, vol
17. Zhang Z, Zhao P, Wang P, Lee W-J (2022) Transfer learning featured short-term combining forecasting model for residential loads with small sample sets. *IEEE Trans Ind Appl* 58(4):4279–4288
18. Ramanan P, Yildirim M, Gebraeel N, Chow E (2023) Differentially private decentralized generator maintenance and Operations for Power Networks. *IEEE Trans Control Netw Syst* 10(2):972–982
19. Han J, Chen Z, Hu P, Li H, Li G, Pi T (2023) IEEE Trans Electr Electron Eng 18(10):1682–1693 Digital Twin Power Grid Oriented Mobile Edge Network Resource Allocation Model,
20. Psaros GN, Tsourakis GI, Papathanassiou SA (2022) Dimensioning of Reactive Power Compensation in an Autonomous Island System, *Applied Sciences-Basel*, vol. 12, no. 6
21. Alencar MV, Silva DN, Nepomuceno L, Martins ACP, Balbo AR, Soler EM (2024) Discrete optimal power flow with prohibited zones, multiple-fuel options, and practical operational rules for control devices. *Appl Energy* 358:122545
22. Yu C et al (2023) Dispatch of highly renewable energy power system considering its utilization via a data-driven bayesian assisted optimization algorithm. *Knowl Based Syst* 281:111059
23. Shah C, King J, Wies RW (2021) Distributed ADMM using private Blockchain for Power Flow Optimization in distribution network with coupled and mixed-integer constraints, *IEEE Access*. 9:46560–46572
24. Wang Z, Zhai C (2022) A distributed compensating strategy for power systems pre-protection against disruptive contingencies using heterogeneous FACTS devices. *Electr Power Syst Res* 213:108630

25. Fu Z, He X, Liu P, Palizban A, Liao W (2022) Distributed Neural Network and Particle Swarm Optimization for micro-grid adaptive power allocation. *Neural Process Lett* 54(4):3215–3233
26. Tung-Lam N, Quoc-Tuan T, Caire R, Wang Y, Besanger Y, Ngoc-An L (2021) Distributed optimal power flow and the multi-agent system for the realization in cyber-physical system. *Electr Power Syst Res* 192:107007
27. Li M, Watson J, Lestas I (2024) Distributed optimal secondary frequency control in Power Networks with Delay Independent Stability. *IEEE Trans Autom Control* 69(6):3748–3763
28. Liu S, Ma J, Zhou T, Zhang Q, Fang W, Liu X (2023) Distributed predictive control design to achieve economically optimal power flow for DC microgrid clusters, *IET Smart Grid*, vol. 6, no. 5, pp. 522–535
29. He Q, Ding L, Kong Z-M, Hu P, Guan Z-H (2022) Distributed Scheme for line overload mitigation with linearized AC Power Flow, *IEEE Transactions on Circuits and Systems II: Express Briefs* 69(6):2877–2881
30. Zheng K, Wang X, Liu J (2022) *IEEE Trans Cloud Comput* 10(2):996–1007 Distributed Traffic Flow Consolidation for Power Efficiency of Large-Scale Data Center Network

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.